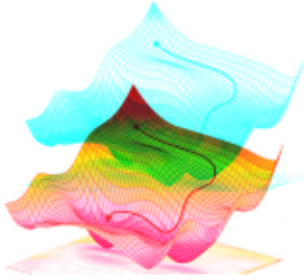
Advanced Risk Minimization

Risk Minimizers



Learning goals

- Know the concepts of the Bayes optimal model (also: risk minimizer, population minimizer)
- Bayes risk
- Consistent learners
- Bayes regret, estimation and approximation error
- Optimal constant model

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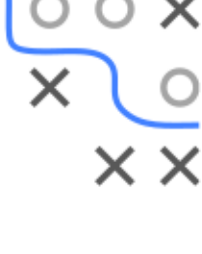
EMPIRICAL RISK MINIMIZATION

Very often, in ML, we minimize the empirical risk



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where

- each observation $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \in \mathcal{X} \times \mathcal{Y}$, so from feature and target space
- $f_{\mathcal{H}} : \mathcal{X} \to \mathbb{R}^g$, $f(\mathbf{x})$ is a model from hypothesis space $\mathcal{H}$; maps a feature vector to output score; sometimes or often we omit $\mathcal{H}$ in the index
- each observation $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{X}, \mathcal{Y}$ are the feature and target spaces, respectively,
- $L : \mathcal{Y} \times \mathbb{R}^g \to \mathbb{R}$ is loss;
- $f : \mathcal{X} \to \mathbb{R}^g$, $f \in \mathcal{H}$ is a model which maps a feature vector to a numerically encoded element of $\mathcal{Y}$ and $\mathcal{H}$ is the hypothesis space,
- $L(y, f(\mathbf{x}))$ measures distance between label and prediction
- We assume that $(\mathbf{x}, y) \sim \mathbb{P}_{xy}$ and $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{xy}$
- $L : (\mathcal{Y} \times \mathbb{R}^g) \to \mathbb{R}$ is the loss function which measures the dissimilarity of the model prediction and the true target,
- $\mathbb{P}_{xy}$ is the distribution of the data generating process (DGP)

Let's define (and minimize) loss in expectation, the theoretical risk:

- and we assume that $\left(\mathbf{x}^{(i)}, y^{(i)}\right) \sim \mathbb{P}_{xy}$ where $\mathbb{P}_{xy}$ is the distribution of the data generating process (DGP).

$$\mathcal{R}(f) := \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] = \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}$$

What is the theoretical justification for this procedure?

TWO SHORT EXAMPLES

Regression with linear model:

- Model: $f(\mathbf{x}) = \boldsymbol{\theta}^\top \mathbf{x} + \theta_0$
- Squared loss: $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$
- Hypothesis space:

$$\mathcal{H}_{\text{lin}} = \left\{ \mathbf{x} \mapsto \boldsymbol{\theta}^\top \mathbf{x} + \theta_0 : \boldsymbol{\theta} \in \mathbb{R}^d, \theta_0 \in \mathbb{R} \right\}$$

Binary classification with shallow MLP:

- Model: $f(\mathbf{x}) = \pi(\mathbf{x}) = \sigma(\boldsymbol{w}_2^\top \text{ReLU}(\boldsymbol{W}_1 \mathbf{x} + \boldsymbol{b}_1) + b_2)$
- Binary cross-entropy loss:
  $L(y, \pi(\mathbf{x})) = -(y \log(\pi(\mathbf{x})) + (1 - y) \log(1 - \pi(\mathbf{x})))$
- Hypothesis space:

$$\mathcal{H}_{\text{MLP}} = \{ \mathbf{x} \mapsto \sigma(\boldsymbol{w}_2^\top \text{ReLU}(\boldsymbol{W}_1 \mathbf{x} + \boldsymbol{b}_1) + b_2) : \mathbf{W}_1 \in \mathbb{R}^{h \times d}, \mathbf{b}_1 \in \mathbb{R}^h, \mathbf{w}_2 \in \mathbb{R}^h, b_2 \in \mathbb{R} \}$$

HYPOTHESIS SPACES AND PARAMETRIZATION

We often write $\mathcal{R}(f)$, but finding an optimal f is operationalized as finding optimal $\theta \in \Theta$ among a set of parametrized curves:

$$\mathcal{H} = \{f_\theta : f_\theta \text{ belongs to certain functional family parametrized by } \theta\}$$

- The params are our means of fixing a specific function from the family. Once set, our model is fully determined.
- This means: finding the optimal model is perfectly equivalent to finding the optimal set of parameter values.
- The relation between optimization over $f \in \mathcal{H}$ and optimization over $\theta \in \Theta$ allows us to operationalize our search for the best model (function) via searching for an optimal θ
- Remark: For some model classes, some parameters encode the same function. So the parameter-to-model mapping could be non-injective (non-identifiability)

OPTIMAL CONSTANTS FOR A LOSS

To derive the risk minimizer observe that by the law of total expectation

$$R(h) = \mathbb{E}_{xy}[L(y, f(x))] = \mathbb{E}_x\left[\mathbb{E}_{y|x}[L(y, f(x)) \mid x = x]\right].$$

We can choose $f(x)$ as we want (unrestricted hypothesis space, non-parametric form).

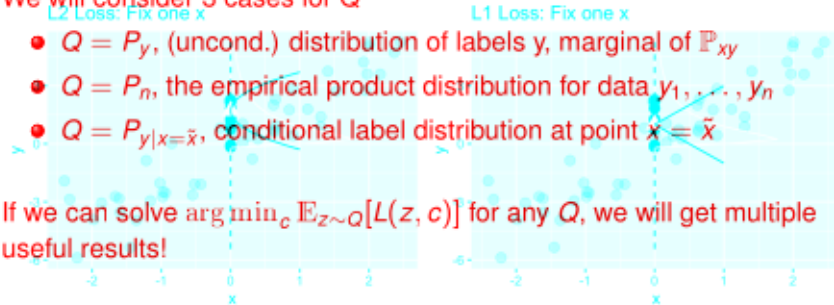
Where for a fixed input x, we choose a constant c we want to predict. So we construct the **point-wise optimizer**

$$f^*(\mathbf{x}) = \text{argmin}_c \mathbb{E}_{y|x}\left[L(y, c) \mid \mathbf{x} = \mathbf{x}\right] \quad \forall \mathbf{x} \in \mathcal{X}.$$

- Let's assume some RV $z \in \mathcal{Y}$ for a label
  - z not the same as y, as we want to fiddle with its distribution
- Assume z has distribution Q, so $z \sim Q$
- We now consider $\arg\min_c \mathbb{E}_{z\sim Q}[L(z, c)]$
- What is the score-constant which approximates z with minimal loss?

We will consider 3 cases for Q

- $Q = P_y$, (uncond.) distribution of labels y, marginal of $\mathbb{P}_{xy}$
- $Q = P_n$, the empirical product distribution for data $y_1, \ldots, y_n$
- $Q = P_{y|x=\tilde{x}}$, conditional label distribution at point $x = \tilde{x}$



If we can solve $\arg\min_c \mathbb{E}_{z\sim Q}[L(z, c)]$ for any Q, we will get multiple useful results!

UNCONDITIONAL: OPTIMAL CONSTANT MODEL

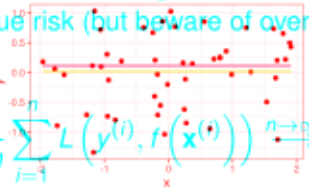
- We would like a loss optimal, constant baseline predictor
- A "featureless" ML model, which always predicts the same value
- Can use it as baseline in experiments, if we don't beat this with more complex model, that model is useless
- Will also be useful as component in algorithms and derivations

The optimal constant model is defined as:

$$f_c^* = \arg\min_{c \in \mathbb{R}} \mathbb{E}_{xy}\left[L(y, c)\right] = \arg\min_{c \in \mathbb{R}} \mathbb{E}_y\left[L(y, c)\right]$$

and $f(\mathbf{x}) = \theta$ is a constant that optimizes the empirical risk $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$ is denoted as as $\hat{f}_c = \arg\min_{c \in \mathbb{R}} \mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$.

Note that according to the **law of large numbers** (LLN), the empirical risk converges to the true risk (but beware of overfitting!):



$$\bar{\mathcal{R}}_{\text{emp}}(f) = \frac{1}{n} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right) \xrightarrow{n \to \infty} \mathcal{R}(f).$$

UNCONDITIONAL: OPTIMAL CONSTANT MODEL

- Let's start with the simplest case, L2 loss
- And we want to find and optimal constant model for

$$\arg\min \mathbb{E}[L(z, c)] =$$

$$\arg\min \mathbb{E}[(z - c)^2] =$$

$$\arg\min \mathbb{E}[z^2] - 2cE[z] + c^2 =$$

$$E[z]$$

- Using $Q \neq P_y$, this means that, given we know the label distribution, the best constant is $c = E[y]$.

RISK MINIMIZER AND APPROXIMATION ERROR / 2

Let us assume we are in an "ideal world":

- The hypothesis space $\mathcal{H} = \mathcal{H}_{all}$ is unrestricted. We can choose any measurable $f : \mathcal{X} \to \mathbb{R}^g$.
- We also assume an ideal optimizer; the risk minimization can always be solved perfectly and efficiently.
- We know $\mathbb{P}_{xy}$.

How should f be chosen?

- $\mathcal{R}_L\left(\hat{f}\right) - \inf_{f \in \mathcal{H}} \mathcal{R}(f)$ is the **estimation error**. We fit $\hat{f}$ via empirical risk minimization and (usually) use approximate optimization, so we usually do not find the optimal $f \in \mathcal{H}$.
- $\inf_{f \in \mathcal{H}} \mathcal{R}_L(f) - \mathcal{R}_L^*$ is the **approximation error**. We need to restrict to a hypothesis space $\mathcal{H}$ which might not even contain the Bayes optimal model f^*.

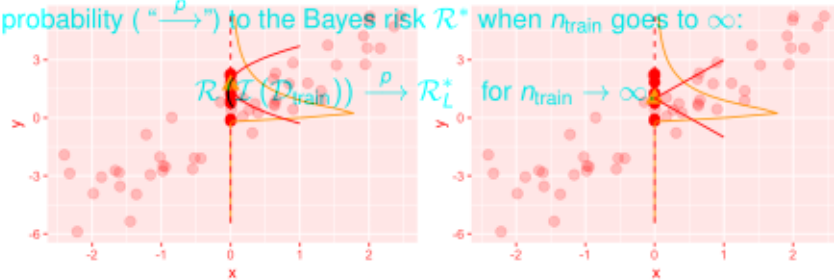
OPTIMAL POINT-WISE PREDICTIONS

To derive the risk minimizer, observe that by law of total expectation

$$\mathcal{R}(f) = \mathbb{E}_{xy}\left[L(y, f(\mathbf{x}))\right] = \mathbb{E}_x\left[\mathbb{E}_{y|x}\left[L(y, f(\mathbf{x}))\,|\,\mathbf{x}\right]\right].$$

- We can choose $f(\mathbf{x})$ as we want (unrestricted hypothesis space, no assumed functional form)
- Hence, for a fixed value $\tilde{\mathbf{x}} \in \mathcal{X}$ we can select **any** value c we want to predict. So we construct the **point-wise optimizer**

$$f^*(\tilde{\mathbf{x}}) = \arg\min_c \mathbb{E}_{y|x}\left[L(y, c)\,|\,\mathbf{x} = \tilde{\mathbf{x}}\right]$$



THEORETICAL AND EMPIRICAL RISK MINIMIZERS / 2

The risk minimizer is mainly a theoretical tool:

- In practice we need to restrict the hypothesis space $\mathcal{H}$ such that we can efficiently search over it.
- In practice we (usually) do not know $\mathbb{P}_{xy}$. Instead of $\mathcal{R}(f)$, we are optimizing the empirical risk

$$\hat{f}_{\mathcal{H}} = \arg\min_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f) = \arg\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

Note that according to the **law of large numbers** (LLN), the empirical risk converges to the true risk (but beware of overfitting!):

$$\bar{\mathcal{R}}_{\text{emp}}(f) = \frac{1}{n} \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right) \overset{n \to \infty}{\longrightarrow} \mathcal{R}(f).$$

ESTIMATION AND APPROXIMATION ERROR

Goal of learning: Train a model $\hat{f}_{\mathcal{H}}$ for which the true risk $\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right)$ is close to the Bayes risk $\mathcal{R}^*$. In other words, we want the **Bayes regret** or **excess risk**

$$\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right) - \mathcal{R}^*$$

to be as low as possible.

The Bayes regret can be decomposed as follows:

$$\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right) - \mathcal{R}^* = \underbrace{\left[\mathcal{R}\left(\hat{f}_{\mathcal{H}}\right) - \inf_{f \in \mathcal{H}} \mathcal{R}(f)\right]}_{\text{estimation error}} + \underbrace{\left[\inf_{f \in \mathcal{H}} \mathcal{R}(f) - \mathcal{R}^*\right]}_{\text{approximation error}}$$

$$= \left[\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}})\right] + \left[\mathcal{R}(f^*_{\mathcal{H}}) - \mathcal{R}(f^*_{\mathcal{H}_{all}})\right]$$